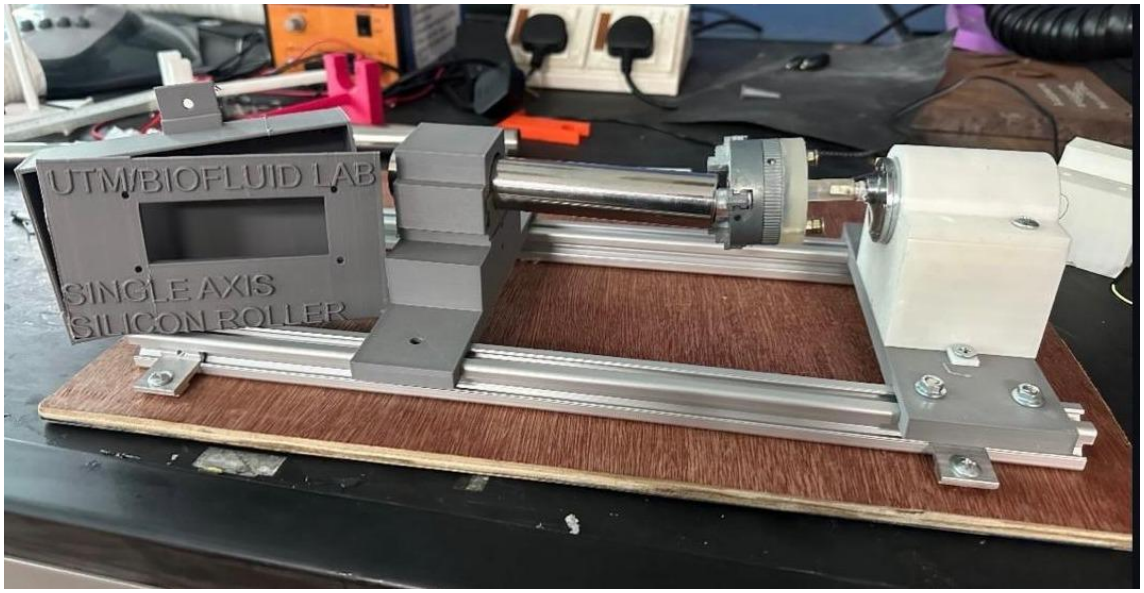
TECHNICAL PROJECT REPORT

Single-Axis Silicone Mould Roller Upgrade

Design, fabrication, testing and documentation of an upgraded silicone mould rotating device for the Biofluid Laboratory



Report number	TR-BFL-RLR-01, Revision 1.0
Prepared for	Dr. Ahmad Zahran Md Khudzari, Senior Lecturer, Biofluid Laboratory (Dr. Zahran Lab)
Prepared by	Mohamed Alamin Ahmed Yousif Ahmed, Bachelor of Electrical Engineering with Honours, FKE
Organisation	Biofluid Laboratory (Dr. Zahran Lab), Department of Biomedical Engineering and Health Sciences, Faculty of Electrical Engineering, Universiti Teknologi Malaysia, 81310 Johor Bahru, Johor
Date	18 October 2026
Related documents	SOP-BFL-RLR-01 Standard Operating Procedure (written and visual)

Executive Summary

The Biofluid Laboratory casts hollow silicone models for flow and device studies by turning a closed mould while the silicone gels. Dr. Ahmad Zahran Md Khudzari requested an upgrade of the laboratory's single-axis silicone mould rotating device into a market-ready unit with better safety, reliability and user interface, together with a detailed standard operating procedure (SOP) and this technical report.

The upgraded roller is built on an aluminium extrusion rail on a plywood base. Every part clamps to the rail with T-nuts, so the bearing support, roller and motor can be slid and re-aligned for different moulds without new parts. A 3D-printed control enclosure, engraved for the laboratory, carries the operator panel with an ON/OFF switch, a speed control and an RPM and status display, and the device is powered from a wall socket. The weakest point of the earlier device, a resin hub that relied only on the fit of its D-bore on the motor shaft, was replaced by a parametrically designed hub disc (Ø47 × 10 mm, 18 mm hub, 6.2 mm D-bore) with an M3 grub screw and captive nut that clamp it to the 6 mm D-shaft. The hub was developed in a browser-based parametric tool with collision warnings and a cut-away view, and refined over several iterations against measurements of the real part.

The roller was assembled and run under power, with the roller tube, chuck, hub and motor aligned on one axis. All seven PDS items are addressed in the design. Two need a final measurement before the device can be declared fully compliant: the speed range up to 100 RPM and the accuracy of the displayed RPM. An acceptance test plan for these is given in Section 6. A written SOP with a one-page visual version has been prepared for training and for display next to the device. Recommendations for market readiness include a chuck guard, a fuse and cable strain relief at the enclosure, a rating and warning label, and a tachometer check of the display.

Table of Contents

Executive Summary 2
1 Introduction 5
1.1 Background 5
1.2 The existing device and the need for an upgrade 5
1.3 Objectives 5
1.4 Scope and deliverables 5
2 Requirements 5
2.1 Product design specification 5
2.2 Implementation guidelines 6
2.3 Constraints 6
3 Design Methodology 6
4 System Design 7
4.1 System architecture 7
4.2 Mechanical layout 7
4.3 Motor seat 8
4.4 Motor-to-roller hub disc 8
4.5 Operator panel and electrical design 10
4.6 Materials and manufacturing 10
4.7 Safety and reliability features 11
5 Fabrication and Assembly 11
5.1 Printing the hub disc 11
5.2 Assembly sequence 11
5.3 Alignment check 11
6 Testing and Results 12
6.1 Results of the build 12
6.2 PDS compliance 12
6.3 Acceptance test plan 12
7 Standard Operating Procedure 13
8 Discussion 14
8.1 Market readiness 14
8.2 Limitations 15
9 Conclusion and Recommendations 15
References 15
Appendix A Reference Speed for Different Mould Radii 16

List of Figures

Figure 1 Functional block diagram of the single-axis silicone mould roller..... 7
Figure 2 Layout of the upgraded single-axis silicone roller with its main parts numbered. 8
Figure 3 CAD model of the 3D-printed motor seat: (a) front, (b) side and (c) underside. 8
Figure 4 Existing resin hub disc used as the reference for the redesign: (a) front, (b) side and (c) back. 9
Figure 5 Development of the hub disc in the parametric 3D tool. 9
Figure 6 CAD model of the redesigned D-bore hub disc: (a) front, (b) back and (c) side. 10
Figure 7 The completed single-axis silicone mould roller. 12
Figure 8 Visual SOP poster for the single-axis silicone mould roller (full size in SOP-BFL-RLR-01, Appendix A). 14

List of Tables

Table 1 Project deliverables 5
Table 2 Product design specification and design response 6
Table 3 Implementation guidelines and how they are met 6
Table 4 Main dimensions of the redesigned hub disc 10
Table 5 Materials and manufacturing processes 10
Table 6 Safety and reliability features of the upgraded roller 11
Table 7 Compliance with the product design specification 12
Table 8 Acceptance tests 12
Table 9 Speed verification record (test T2) 13
Table 10 Market-readiness assessment 15
Table 11 Reference speed $n_c = 29.9 / \sqrt{r}$ for different mould radii 16

1 Introduction

1.1 Background

Silicone models of blood vessels and organs are widely used in biofluid and cardiovascular research for flow visualisation, testing of medical devices and training. Hollow silicone parts of this kind are often made by rotational casting: a measured amount of liquid silicone is poured into a closed mould, and the mould is turned so that the silicone coats the inner wall while it cures. The result is a hollow part with a controlled wall thickness, made without a core. The quality of the part depends strongly on the motion. The mould must turn smoothly, at a steady and repeatable speed, for at least the gel time of the silicone.

For tubular moulds such as straight or gently curved vessel models, rotation about a single axis along the mould is enough to spread the silicone around the circumference. This is the job of the single-axis silicone mould roller.

1.2 The existing device and the need for an upgrade

The laboratory already had a single-axis silicone mould rotating device. It worked, but it was not a finished product: the connection between the motor and the roller relied on a translucent resin hub that was held on the motor's D-shaft only by the fit of its bore, so it could work loose along the shaft. The project brief asked for the device to be replaced with a market-ready solution with improved safety, reliability and user interface, supported by complete documentation and operating guidelines.

1.3 Objectives

(1) To design and fabricate a single-axis silicone mould roller that meets the product design specification (PDS) issued by the laboratory.

(2) To provide a safe and reliable mechanical drive, in particular a secure coupling between the motor shaft and the roller chuck.

(3) To provide a simple operator panel with ON/OFF control, speed control and an RPM and status display, powered from a wall socket.

(4) To document the device with a detailed SOP in written and visual (infographic) formats and with this technical report.

1.4 Scope and deliverables

Table 1 Project deliverables

No.	Deliverable (from the project brief)	Where it is delivered
1	Design and fabricate a single-axis silicone mould roller device	The device in the Biofluid Laboratory; design described in Sections 3 to 5
2	Draft detailed Standard Operating Procedures (SOP)	SOP-BFL-RLR-01, with the visual SOP poster in its Appendix A; summarised in Section 7
3	Prepare a comprehensive technical project report	This report, TR-BFL-RLR-01

2 Requirements

2.1 Product design specification

Table 2 lists the PDS items from the project brief, how each was interpreted in the design and how it is verified.

Table 2 Product design specification and design response

ID	PDS requirement	Design response	Verification
a	Single-axis rotational mechanism	Mould carried on a stainless-steel roller tube turning about one horizontal axis, driven through a chuck and hub disc	Inspection; run test
b	Smooth operation at speeds up to 100 RPM	Motor in line with the roller axis on a cradle seat; positive shaft clamp; speed set from the panel	Tachometer at set points (test T2); run-out and smoothness check (T3)
c	Intuitive speed control interface	Single speed control knob on the operator panel, next to the display	User trial (T5)
d	Wall-socketted power supply	Wall-socket adapter feeding the control enclosure	Inspection
e	ON/OFF control via dedicated operator panel	ON/OFF switch on the front of the control enclosure	Function test (T5)
f	LED display for RPM and operational status	Display mounted behind the window of the control enclosure	Display against tachometer (T2)
g	3D-printed or equivalent materials	PLA motor seat, bearing support, hub disc and enclosure; aluminium rail; plywood base	Inspection

2.2 Implementation guidelines

Table 3 Implementation guidelines and how they are met

Guideline from the brief	How it is addressed
Focus on user-friendly interface design	Three controls only (ON/OFF, speed, display) grouped on one engraved panel; rail layout keeps the roller open and easy to load
Prioritise safety and reliability in all mechanical and electrical systems	Positive clamp on the motor shaft; adjustable, re-alignable rail mounting; wall-socket adapter so the enclosure is not at mains voltage; hazard analysis and controls in the SOP
Document all procedures with clear illustrations and step-by-step instructions	SOP with labelled photographs, step tables with acceptance checks, troubleshooting and forms
Include written and visual (infographic) SOP formats	Written SOP-BFL-RLR-01 and a one-page colour visual SOP poster

2.3 Constraints

- Compatibility with the laboratory's existing moulds and with the existing drive motor and its 6 mm D-shaft, which the new hub had to fit.
- Parts to be produced on the laboratory's 3D printers or bought from local online suppliers.
- Bench-top size, safe for use in a shared teaching and research laboratory.

3 Design Methodology

The device was developed in five stages. First, the brief and the existing device were reviewed and the PDS was agreed with the supervisor. Second, the layout was planned around an extrusion rail so that the existing roller tube, chuck and motor could be mounted and aligned. Third, the parts that had to be made were modelled before printing: the motor seat, the bearing support, the enclosure and the hub disc. The hub disc, being the critical part, was modelled parametrically and checked against measurements of the real part after each iteration. Fourth, the parts were printed and assembled, and the drive line was aligned. Fifth, the device was run under power and the SOP and this report were written from the as-built device. Whenever a dimension was found to be wrong, the model was corrected and the part reprinted, so that the design files always match the hardware.

4 System Design

4.1 System architecture

Figure 1 shows the functional blocks of the roller. Power from the wall socket passes through the adapter and the ON/OFF switch to the speed controller in the control enclosure, which drives the motor at the speed set by the knob and shows the RPM and status on the display. The motor turns the roller through the hub disc and chuck, and the roller carries the mould.

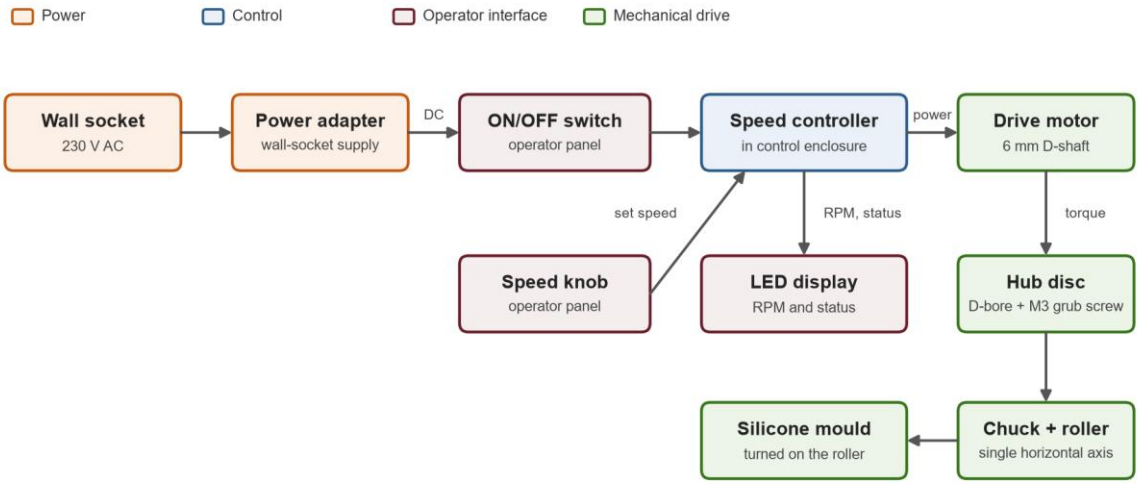
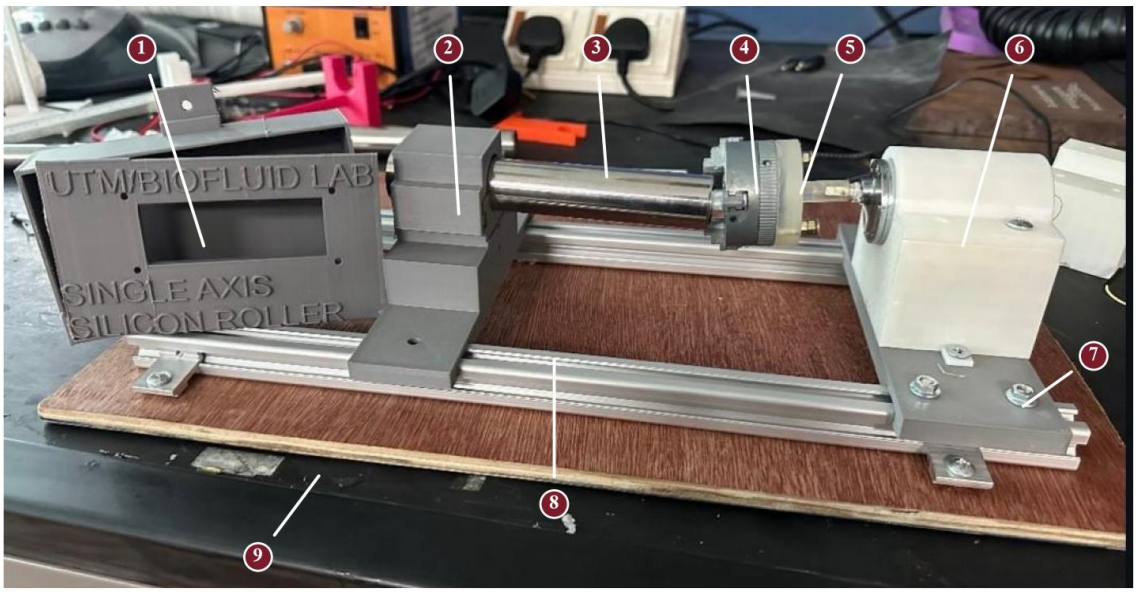


Figure 1 Functional block diagram of the single-axis silicone mould roller.

4.2 Mechanical layout

The device is built on a plywood base carrying an aluminium extrusion rail (Figure 2). All other components clamp to the rail with T-nuts, so their positions along the roller axis can be adjusted to suit different moulds and re-aligned without new parts. From left to right, the device consists of the 3D-printed control enclosure, a 3D-printed bearing support carrying a polished stainless-steel roller tube, a knurled chuck at the driven end of the tube, the hub disc, and the motor housing on its own bracket. The control enclosure houses the controller and display behind a rectangular window and is engraved UTM/BIOFLUID LAB, SINGLE AXIS SILICON ROLLER.



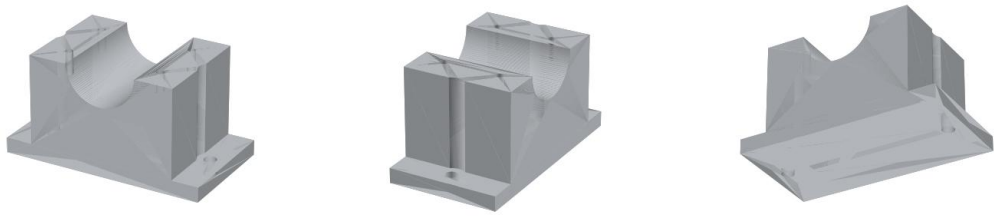
1 Control enclosure with display window
2 3D-printed bearing support
3 Stainless-steel roller tube
4 Knurled chuck
5 D-bore hub disc
6 Motor housing
7 Motor bracket with T-nut clamps
8 Aluminium extrusion rail
9 Plywood base

Figure 2 Layout of the upgraded single-axis silicone roller with its main parts numbered.

Because the roller turns about one axis only, its load is simple. The mould and tube are supported on the axis, so the motor mainly overcomes bearing friction and the small torque from any offset of the mould and its liquid contents. The design effort was therefore spent on alignment and on a secure drive connection rather than on torque.

4.3 Motor seat

The motor is carried by a 3D-printed motor seat (Figure 3). The seat is a 79 × 59 mm block, 56 mm high, with a semicircular cradle that supports the round motor body and a flanged base with fixing holes. Holding the motor body in a cradle keeps the motor shaft in line with the roller axis, which reduces the side load on the hub and the chuck and helps the roller run smoothly.



(a) Front: cradle for the motor body (b) Side: flanged base with fixing holes (c) Underside

Figure 3 CAD model of the 3D-printed motor seat: (a) front, (b) side and (c) underside.

4.4 Motor-to-roller hub disc

4.4.1 Problem

The motor has a 6 mm D-shaped output shaft, and the chuck needs a flange to bolt to. The existing translucent resin hub (Figure 4) relied only on the fit of its D-bore. A D-bore transmits torque well, but

nothing stopped the hub from sliding along the shaft, and as the bore wore the fit became loose. A loose hub lets the roller lag, knock or stop, which spoils the casting and can let the mould shift.



Figure 4 Existing resin hub disc used as the reference for the redesign: (a) front, (b) side and (c) back.

4.4.2 Parametric development

A new hub was modelled from the existing part and given a set screw. It was developed in a parametric, browser-based 3D tool built for this task, with a slider for every dimension, live warnings when two features collide, a cut-away view and a button that exports the STL file. The first model was scaled from photographs, which gave a disc of about Ø55 × 14 mm with eight ring segments on the back. Measurements of the real part then corrected it step by step (Figure 5): the back ring was reduced to four segments with a 3.5 mm edge band, the mounting holes were placed with their outer edge 3.5 mm from the disc edge, the motor shaft was confirmed as a 6 mm D-type, the body was set to 10 mm with a Ø29 mm hollow 5.5 mm deep, and the hub was raised to 20 mm. Each change was checked in the cut-away view before the STL was exported.

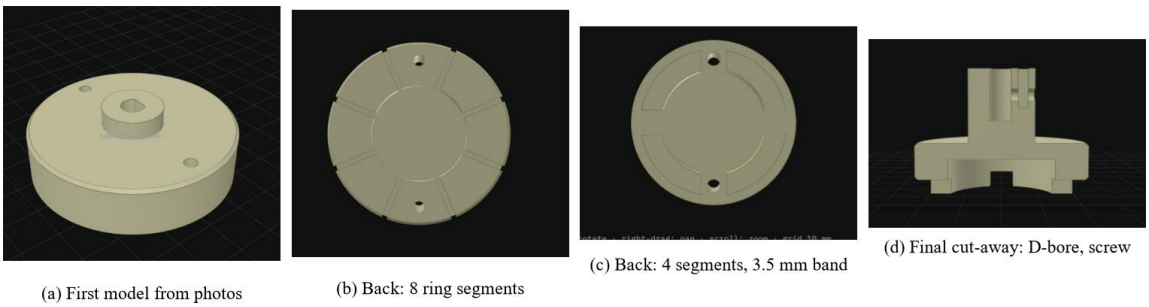
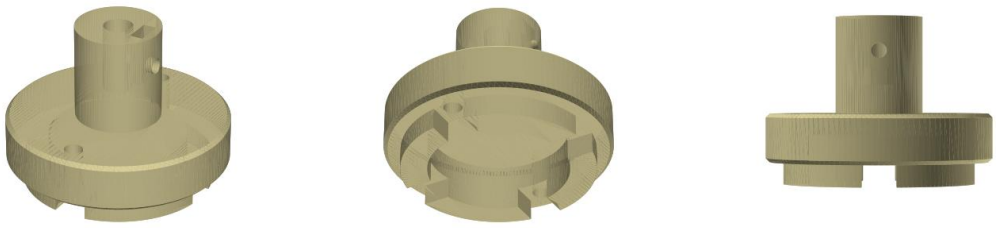


Figure 5 Development of the hub disc in the parametric 3D tool.

4.4.3 Final design

The final disc is 47 mm in diameter and 10 mm thick, with an 18 mm hub standing 20 mm above the front face (Figure 6 and Table 4). The hub carries a 6.2 mm D-bore with a 5.7 mm flat, which gives a sliding fit on the 6 mm D-shaft (5.5 mm across the flat) and allows for the slight undersize of printed holes. An M3 grub screw, fed from a nut slot in the side of the hub, clamps against the flat of the shaft, so the hub can neither slip round nor slide off. On the back, four ring segments 4.5 mm tall locate the disc on the chuck around the Ø29 mm hollow, and two Ø3.8 mm holes take the screws to the chuck.



(a) Front: Ø18 hub with 6 mm D-bore (b) Back: segmented ring and Ø29 hollow (c) Side view

Figure 6 CAD model of the redesigned D-bore hub disc: (a) front, (b) back and (c) side.

Table 4 Main dimensions of the redesigned hub disc

Feature	Dimension
Disc body	Ø47 mm × 10 mm, 0.8 mm edge chamfer
Overall height	34.5 mm
Front hub	Ø18 mm × 20 mm above the front face
Motor shaft	Ø6 mm D-type, 5.5 mm across the flat
D-bore	Ø6.2 mm, 5.7 mm across the flat (0.2 mm clearance on both)
Shaft clamping	M3 × 6 grub screw with captive M3 nut in a side slot
Back ring	4 segments, 4.5 mm tall, 3.5 mm edge band
Centre hollow	Ø29 mm, 5.5 mm deep into the body
Mounting holes	2 × Ø3.8 mm on a Ø36.2 mm pitch circle
Volume and mass	19.86 cm³ (about 24.6 g if printed solid in PLA)

4.5 Operator panel and electrical design

The operator panel follows the PDS directly: one ON/OFF switch, one speed control and one display that shows the RPM and the operating status. The three are grouped on the front of the control enclosure, so a user can start, set and monitor a run from one place, and the panel is engraved with the laboratory and device name so it can be identified in a shared laboratory. Power comes from a wall-socket adapter, so the enclosure and the motor work at low DC voltage and no mains wiring is exposed inside the device. Unplugging the adapter isolates the device for mounting, cleaning and maintenance, and this is the isolation step used throughout the SOP.

The enclosure is placed at the opposite end of the rail from the motor and mould drive, so that a spill from the mould is unlikely to reach it, and it is mounted on the rail like the other parts so it stays square to the roller.

4.6 Materials and manufacturing

Table 5 Materials and manufacturing processes

Part	Material	Process
Base	Plywood	Cut to size
Mounting rail	Aluminium extrusion with T-slots	Cut to length; parts fixed with T-nuts and screws
Roller tube	Polished stainless steel	Standard part
Chuck	Metal, knurled	Standard part
Hub disc	PLA	3D printed from the parametric model (Section 5.1)
Motor seat, bearing support, motor bracket	PLA	3D printed
Control enclosure with engraved panel	PLA	3D printed, text engraved in the model

Part	Material	Process
Fasteners	Steel M3 grub screw and nut; T-nuts and screws	Bought in

4.7 Safety and reliability features

Table 6 Safety and reliability features of the upgraded roller

Feature	Benefit
Grub screw clamp on the D-flat with a captive nut	Hub cannot slide off or slip; the drive no longer depends on the fit of the bore alone
Motor body held in a cradle seat in line with the roller	Less side load on the hub and chuck; smoother running and less wear
All parts clamped to one extrusion rail	Parts stay aligned and square; positions can be adjusted for new moulds without new parts
Wall-socket adapter	Low-voltage DC inside the device; simple isolation by unplugging
Single grouped operator panel	Fewer operating errors; speed can be turned down before switching on and off
Parametric hub model	A worn or broken hub, or one for a different motor or chuck, can be reprinted in minutes
Written and visual SOP	Consistent use by different users; hazards and emergency actions shown next to the device

5 Fabrication and Assembly

5.1 Printing the hub disc

The hub is printed back-face down, so that the ring segments sit on the bed and the hub points up. The roof of the centre hollow is then a Ø29 mm bridge, so supports are switched on if the printer sags on it. The bore is printed vertically, which gives the roundest hole and keeps the flat accurate.

5.2 Assembly sequence

1. Fix the aluminium extrusion rail to the plywood base.
2. Slide T-nuts into the rail for the control enclosure, the bearing support and the motor bracket.
3. Mount the motor in its seat and housing, and fix the bracket loosely to the rail.
4. Drop the M3 nut into the slot on top of the hub before the shaft is inserted. Slide the hub onto the D-shaft with the bore flat aligned to the shaft flat, then turn the grub screw in from the side until it bears on the flat.
5. Fit the chuck and roller tube, and place the free end of the tube in the bearing support.
6. Slide the bearing support and motor bracket along the rail until the roller runs true when turned by hand, then tighten their T-nut screws.
7. Tighten the two screws between the hub and the chuck last, after checking again that the roller runs true.
8. Mount the control enclosure, route the motor and adapter cables away from the roller, and close the enclosure.

5.3 Alignment check

With the device switched off, the roller is turned slowly by hand through several turns. It must turn freely, without tight spots, rubbing or visible wobble at the chuck. If it binds, the bearing support or motor bracket is moved slightly along or across the rail and re-tightened. This check is repeated in the SOP before every run.

6 Testing and Results

6.1 Results of the build

The completed roller is shown in Figure 7. All components are mounted on the extrusion rail on a common plywood base, the control enclosure is engraved for the laboratory, and the roller tube, chuck, hub and motor are aligned on one axis. The device was assembled and run under power. Because every component clamps to the rail, the bearing support and motor can be moved and re-aligned without new parts. The redesigned hub adds a positive clamp on the flat of the motor shaft, so the drive no longer depends on the fit of the D-bore alone, and its parametric model allows a new hub to be printed quickly if a different motor or chuck is used.

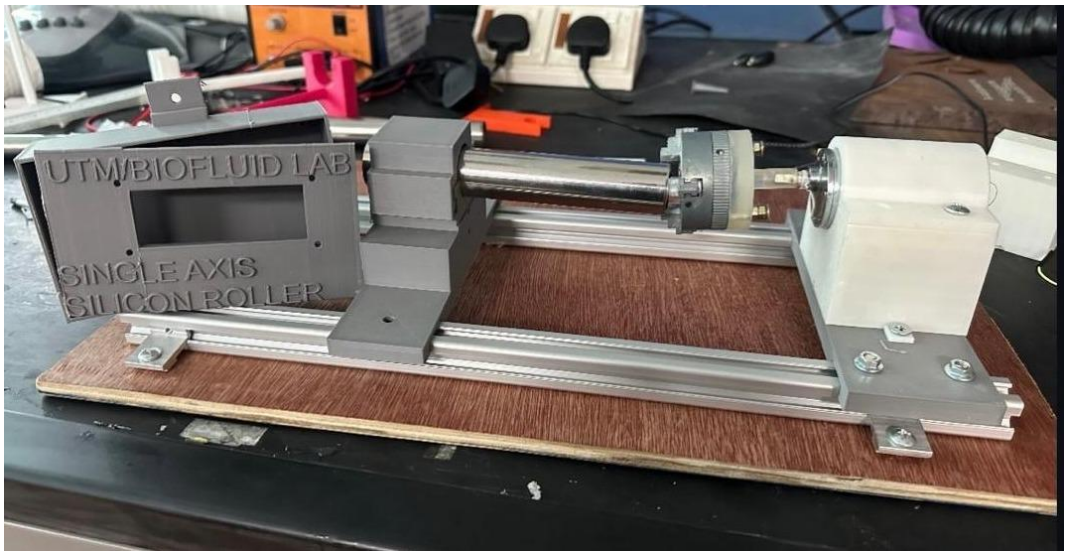


Figure 7 The completed single-axis silicone mould roller.

6.2 PDS compliance

Table 7 Compliance with the product design specification

ID	Requirement	Status	Evidence or remaining action
a	Single-axis rotation	Met	Roller turns about one horizontal axis (Figure 7)
b	Smooth operation up to 100 RPM	Built; to verify	Runs smoothly under power. Measure the top speed and smoothness with a tachometer (T2, T3)
c	Intuitive speed control	Met	Single knob on the operator panel
d	Wall-socket power supply	Met	Wall-socket adapter
e	ON/OFF on a dedicated operator panel	Met	ON/OFF switch on the control enclosure
f	LED display of RPM and status	Met; accuracy to verify	Display fitted behind the enclosure window. Compare with a tachometer (T2)
g	3D-printed or equivalent materials	Met	PLA printed parts on an aluminium rail and plywood base

6.3 Acceptance test plan

The following tests complete the verification of the device before it is released for routine use. They need only a handheld optical tachometer, a stopwatch, a calliper and a representative mould. Results are recorded in Table 9 and in the device folder.

Table 8 Acceptance tests

Test	Purpose	Method	Pass criterion
T1	Mechanical inspection	Pre-use inspection of SOP 9.1; check every clamp and screw	All checks pass

Test	Purpose	Method	Pass criterion
T2	Speed range and display accuracy	Reflective tape on the chuck; measure speed with the tachometer at minimum, 25 %, 50 %, 75 % and maximum knob setting; record the displayed RPM	Maximum speed ≥ 100 RPM; display within ±5 % of measured
T3	Smoothness	At each speed in T2, watch the chuck and mould for wobble and listen for knocking	No visible wobble or knocking
T4	Endurance	Run for 60 minutes at the highest speed used for casting with a representative filled mould	No slip, no loosening, no abnormal heating or noise
T5	Controls and user trial	A new user switches on, sets a speed, reads the display and switches off using only the visual SOP	Completed without help or error
T6	Casting trial	Cast a tubular part following the SOP; measure the wall at 6 points around and along it with a calliper	Wall thickness within the tolerance set for the model

Table 9 Speed verification record (test T2)

Knob setting	Displayed RPM	Measured RPM	Error (%)	Smooth? (Y/N)
Minimum				
25 %				
50 %				
75 %				
Maximum				

7 Standard Operating Procedure

The SOP, document SOP-BFL-RLR-01, was written from the as-built device and follows the implementation guidelines of the brief. It contains the purpose and scope, responsibilities, a labelled description of the device and its controls, a hazard and control table with PPE requirements, step-by-step procedures with an acceptance check for every step (pre-use inspection, silicone preparation, mould mounting, start-up and speed setting, running, shutdown and demoulding, cleaning), emergency actions, a maintenance schedule, a troubleshooting guide, a pre-use checklist and run log, a training sign-off and a hub replacement procedure.

For the visual format, a one-page colour poster (Figure 8) summarises the PPE, the hazards, the four phases of a run (prepare, mount mould, run, stop and clean), the main do's and don'ts and the emergency actions. It is intended to be printed on A4 and displayed next to the device, and used in training together with the written SOP.



Figure 8 Visual SOP poster for the single-axis silicone mould roller (full size in SOP-BFL-RLR-01, Appendix A).

To help users choose a speed, the SOP gives a reference speed at which the centrifugal acceleration at the inner wall of the mould equals gravity:

$$n_c = (60 / 2\pi)\sqrt{(g / r)} \approx 29.9 / \sqrt{r} \quad (1)$$

where n_c is in RPM, g = 9.81 m/s^2 and r is the inner radius of the mould cavity in metres. Below n_c the silicone tumbles over the wall; well above it the silicone is held against the wall as an even film around the circumference but cannot flow along the mould. For a mould radius of 10 to 100 mm, n_c falls from about 299 to 95 RPM, so the 100 RPM PDS target covers both regimes for typical vessel moulds.

8 Discussion

8.1 Market readiness

The brief asked for a market-ready solution. Table 10 assesses the device against the main aspects of a laboratory product and lists what remains to be done.

Table 10 Market-readiness assessment

Aspect	Status	Action to reach market readiness
Function	Device built and run under power	Complete tests T2 to T6 and record the results
Mechanical reliability	Positive shaft clamp; aligned rail layout	Monitor hub wear in the maintenance log; keep a spare printed hub
Electrical safety	Low-voltage DC from a wall adapter	Add an inline fuse and cable strain relief where the cables enter the enclosure; confirm the adapter is a certified unit
Guarding	Rotating chuck and hub are exposed	Add a clear printed or acrylic guard over the chuck and hub, open only at the mould
Labelling	Panel engraved with lab and device name	Add a rating label (voltage, current), a rotating-parts warning label and the SOP number
Speed indication	Display fitted	Verify against a tachometer (T2); recalibrate if outside ±5 %
Documentation	Written SOP, visual SOP and this report	Review after the first month of use
Reproducibility	Parametric hub model and printed parts	Archive the CAD and STL files with a bill of materials in the laboratory drive

8.2 Limitations

- The speed range and display accuracy have not yet been measured with an independent instrument.
- The chuck and hub are not guarded; the SOP controls this risk by procedure only.
- PLA parts can creep under sustained load or warm conditions. The hub and supports should be inspected monthly as set out in the SOP.
- The roller is built around the current roller tube and chuck; a different mould diameter may need a new chuck or adaptor.

9 Conclusion and Recommendations

The single-axis silicone mould roller was upgraded into a tidy, adjustable bench device on an aluminium extrusion rail, with a labelled control enclosure carrying an ON/OFF switch, a speed control and an RPM and status display, powered from a wall socket. The weak motor coupling of the earlier device was replaced by a parametrically designed hub disc with a grub screw clamp, which makes the drive positive and easy to replace. The device was assembled and run under power, and it meets the PDS in its design. A detailed SOP in written and visual formats was prepared, completing the three deliverables of the brief.

The following actions are recommended:

1. Carry out acceptance tests T2 to T6 and record the results in Table 9 before routine use.
2. Fit a clear guard over the chuck and hub, an inline fuse and cable strain relief, and rating and warning labels.
3. Archive the CAD and STL files of all printed parts, with this report and the SOP, on the laboratory drive.
4. Record the speed that gives the best result for each mould on its casting sheet, building a reference for future users.
5. Review the SOP after one month of use and update it with user feedback.

References

Crawford, R. J., & Throne, J. L. (2002). Rotational molding technology. William Andrew Publishing.

Budynas, R. G., & Nisbett, J. K. (2015). Shigley's mechanical engineering design (10th ed.). McGraw-Hill Education.

Hibbeler, R. C. (2016). Engineering mechanics: Dynamics (14th ed.). Pearson.

IEC 61010-1:2010+A1:2016. Safety requirements for electrical equipment for measurement, control, and laboratory use. Part 1: General requirements. International Electrotechnical Commission.

Department of Biomedical Engineering & Health Sciences. (n.d.). Faculty of Electrical Engineering, Universiti Teknologi Malaysia. https://fke.utm.my/biomedical/

Appendix A Reference Speed for Different Mould Radii

Table 11 Reference speed $n_c = 29.9 / \sqrt{r}$ for different mould radii

Mould inner radius r (mm)	10	20	30	50	75	100
Reference speed n_c (RPM)	299	211	173	134	109	95